
Applied Project Proposal: Hybrid Movie Recommendation System with Cold-Start Support

CS4210 – Machine Learning and Its Applications

Project type: Applied project	Goal: build a working recommendation prototype, compare ML baselines, and demo personalized results
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Project idea. I propose to build a **hybrid movie recommendation system** that produces personalized top-N suggestions after a user rates a small set of movies. The core machine learning challenge is learning useful preferences from a sparse user-item matrix while still handling **cold-start** items that have little rating history. This is interesting because recommendation is a realistic product problem: the system must rank items well, remain useful with limited data, and generate explanations that make its behavior understandable. It is also a good fit for this course project because it is meaningfully ML-based yet different from the uploaded lecture set, which focuses on linear models and neural-network/CNN vision topics rather than recommender systems.



Related work / GitHub references. The main dataset will be **MovieLens 100K or 1M**. For implementation references and baselines, I will study **Surprise** for explicit-rating collaborative filtering, **LightFM** for hybrid latent-factor recommendation with ranking losses, and **RecBole** for experiment organization and evaluation. Methodologically, the project is informed by classic **matrix factorization** for recommender systems and pairwise ranking ideas such as **BPR**. These references give me a clear path to build a system that is both feasible and technically meaningful.

Proposed methods / pipeline. I will combine **user-item interactions** with **movie metadata** such as genres and tags. After cleaning and encoding users/items, I will create train, validation, and test splits, plus a smaller **cold-start split** in which some items are held out from collaborative training. I will compare three levels of models: (1) a popularity baseline, (2) an item-item collaborative filtering or SVD baseline, and (3) the proposed **hybrid latent-factor model** that uses both interaction signals and content features. The main evaluation will focus on recommendation quality using **Precision@10**, **Recall@10**, **NDCG@10**, and **catalog coverage**; if I keep explicit rating prediction in the pipeline, I will also report **RMSE/MAE**. An ablation study will compare collaborative-only versus hybrid features to measure the value of metadata.

System components snapshot.

System component	Role in the final prototype
Data layer	Load MovieLens ratings and movie metadata, encode users/items, and create standard plus cold-start splits.
Model layer	Train popularity, item-CF/SVD, and the proposed hybrid latent-factor recommender.
Evaluation layer	Report Precision@10, Recall@10, NDCG@10, coverage, and optional RMSE/MAE.
Inference / UI	Collect seed ratings from a user and return personalized recommendations with short explanations.

Planned implementation. The system will be built in **Python** with pandas/NumPy for preprocessing, Surprise for classical baselines, and LightFM or a small **PyTorch** implementation for the hybrid model. The codebase will be organized into modules for data preparation, training, evaluation, saved artifacts, and inference. I also plan to add a simple explanation layer that surfaces “**because you liked ...**” style reasons using nearest neighbors in embedding space and shared genres/tags. The final submission will include reproducible scripts/notebooks, result tables or plots, and a lightweight app layer for inference.

Planned demo. In the final demo, a peer will rate about **5–10 movies** or choose a few favorites, and the prototype will generate personalized recommendations in real time. I expect to show four things: the end-to-end system pipeline, a side-by-side comparison between popularity and personalized recommendations, a cold-start example for a new movie that relies on metadata, and a compact results dashboard with ranking metrics and example explanations. This should make the machine learning component, system design, and product behavior easy to present to classmates.

Backup direction (if revision is needed). If recommendation systems are later scheduled as a course topic, I can pivot to a **graph-based fraud detection** project on transaction data while keeping the same product-style prototype and evaluation structure.

Selected references. MovieLens (GroupLens); Surprise (NicolasHug/ Surprise); LightFM (lyst/lightfm); RecBole (RUCAIBox/RecBole); Koren, Bell, and Volinsky, “Matrix Factorization Techniques for Recommender Systems”; Rendle et al., “BPR: Bayesian Personalized Ranking from Implicit Feedback.”